

Andrew Liu

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EDUCATION

University of California, Santa Barbara
College of Engineering - B.S. Electrical Engineering
Expected Graduation: June 2028
Cumulative GPA: 3.98/4.00

Relevant Coursework:

Circuits and Systems (Analog and Digital) I-III + Laboratories I-III
Fundamentals of Logic Design, Digital Design Principles, Digital Image Processing
Physics: Electricity and Magnetism, Optics and Photonics, Modern Physics + Laboratory
Computer Science: Data Structures and Algorithms, Object-Oriented Programming (C++)

WORK EXPERIENCE

Electrical Engineering Intern | Hoplynk June 2026 - Present

- Designed and debugged a Power Distribution and Battery Monitoring Printed Circuit Board
- Ensured EMI from high-power switching areas does not inflict EMI onto sensitive analog components
- Ran simulations on LTSpice to simulate battery charge/discharge behavior and locate points of failure
- Assembled and tested various electronics using a DMM and running test software (e.g., radios)

Avionics/Electrical Team Lead | Gaucho Rocket Project January 2026 - Present

- Select compatible connectors for MCU-to-sensor connections by reading blueprints and datasheets
- Verify electronic components and sensor functionality by taking measurements with test equipment
- Document test results and create protocols so the team has reference data and test instructions

PROJECTS

Gaucho Rocket Project Sensor Logger Printed Circuit Board (PCB Design) April 2026 - May 2026

- Designed a 4-layer PCB in KiCad to log temperature, pressure, acceleration data during rocket flight
- Selected components accounting for size, power, and sensor compatibility
- Validated and tested key components and sensors (e.g., thermocouple amplifier) on breakout boards

Boost Converter Printed Circuit Board (KiCad, Soldering, Power Electronics) March 2026 - April 2026

- Designed and prototyped a boost converter circuit, achieving an efficiency of ~66% (5V to 12V)
- Implemented pulse width modulation control with closed-loop voltage feedback using Arduino/ATtiny
- Designed PCB layout, taking into account trace width calculations and component placement

Plantar Fasciitis Risk Screening Website (JavaScript, MediaPipe) January 2026

- For the Santa Barbara Hackathon, helped develop a website to screen for risk of foot condition
- Modeled ankle, knee, heel joints in the vector space and formulated a benchmark 5-degree threshold

Arduino Remote Control Car (Circuits, CAD, C++) August 2025 - September 2025

- Designed and built remote-controlled car using a microcontroller and other electronic components
- Programmed using Arduino IDE to handle motor control, steering, and wireless communication
- Used CAD software and 3D printing software to design and make a chassis for mounting components

SKILLS

Technical Skills: Analog and Digital Circuit Analysis, Circuit prototyping and debugging, Soldering, PCB Design (KiCad), C++, Python, Oscilloscope, Multimeter, Function Generator, Datasheets, LTSpice